Mark schemes

Q1.

(a)

Marking Instructions	AO	Marks	Typical Solution
Demonstrates the rule is correct for $n = 1$	AO1.1b	B1	$\sum_{r=1}^{1} r^3 = 1^3 = 1$ and $\frac{1}{4} \times 1^2 \times 2^2 = 1$ ∴ it is true for $n = 1$
States the rule is true for $n = k$ **and** adds $(k + 1)^3$ to $\frac{1}{4}k^2(k+1)^2$	AO2.4	M1	Assume it it true for $n = k$ $\sum_{r=1}^{k} r^3 = \frac{1}{4}k^2(k+1)^2$
Obtains $\frac{1}{4}(k+1)^2(k+2)^2$ from $\frac{1}{4}k^2(k+1)^2 + (k+1)^3$	AO2.2a	A1	$\sum_{r=1}^{k} r^3 + (k+1)^3 = \frac{1}{4}k^2(k+1)^2 + (k+1)^3$
Completes a rigorous argument and explains how their argument proves the required result. This mark is only available if all previous marks have been awarded.	AO2.1	R1	$\sum_{r=1}^{k+1} r^3 = \frac{1}{4}(k+1)^2(k^2 + 4(k+1))$ $\sum_{r=1}^{k+1} r^3 = \frac{1}{4}(k+1)^2(k^2 + 4k + 4)$ $\sum_{r=1}^{k+1} r^3 = \frac{1}{4}(k+1)^2(k+2)^2$ ∴ it is also true for $n = k + 1$ True for $n = 1$, and true for $n = k \Rightarrow$ true for $n = k + 1$, then by induction it is true for all integers $n \geq 1$ **AG**

(b)

Marking Instructions	AO	Marks	Typical Solution
Expresses LHS as summations of r^3 and r Ignore limits of the sums in this part only.	AO1.1b	B1	$\sum_{r=1}^{2n} r(r-1)(r+1) = \sum_{r=1}^{2n} (r^3 - r)$ $= \sum_{r=1}^{2n} r^3 - \sum_{r=1}^{2n} r$
Expresses LHS in terms of n, using part (a) and $\sum r = \frac{1}{2}n(n+1)$	AO1.1a	M1	$= \frac{1}{4}(2n)^2(2n+1)^2 - \frac{1}{2}2n(2n+1)$
Takes out $n(2n + 1)$ as a factor or obtains $4n^4 + 4n^3 - n^2 - n$ Allow one slip in second bracket or one incorrect term in the expansion.	AO1.1a	M1	$= n^2(2n + 1)^2 - n(2n + 1)$ $= n(2n + 1)(n(2n + 1) - 1)$
Completes fully correct proof to reach the required result. This mark is only available if all previous marks have been awarded.	AO2.1	R1	$= n(2n + 1)(2n^2 + n - 1)$ $= n(2n + 1)(2n - 1)(n + 1)$

Note: $n(n+1)(2n-1)(2n+1) = 4n^4 + 4n^3 - n^2 - n$			$= n(n+1)(2n-1)(2n+1)$ **AG**
			Total 8 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Uses proof by induction and investigates formula for $n = 1$ and $n = k$ (must see evidence of both $n = 1$ and $n = k$ being considered)	AO3.1a	M1	Using induction method, Let P(n) be the statement $\mathbf{M}^n = \begin{bmatrix} 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \end{bmatrix}$ For $n = 1$
Demonstrates that formula is true for $n = 1$	AO1.1b	A1	$\begin{bmatrix} 3^0 & 3^0 & 3^0 \\ 3^0 & 3^0 & 3^0 \\ 3^0 & 3^0 & 3^0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \mathbf{M}^1$
States assumption that formula true for $n = k$ and uses $\mathbf{M}^{k+1} = \mathbf{M} \times \mathbf{M}^k$	AO2.1	R1	$\therefore$P(1) is true Assume P(k) is true $\mathbf{M}^{k+1} = \mathbf{M} \times \mathbf{M}^k$
Deduces that formula is also true for $n = k + 1$ from correct working	AO2.2a	R1	$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \end{bmatrix}$ (since P(k) is true)
Completes a rigorous argument and explains how their argument proves the required result. **AG**	AO2.4	R1	$= \begin{bmatrix} (3^{k-1}+3^{k-1}+3^{k-1}) & (3^{k-1}+..) & (3^{k-1}+.. \\ (3^{k-1}+3^{k-1}+3^{k-1}) & (3^{k-1}+..) & (3^{k-1}+.. \\ (3^{k-1}+3^{k-1}+3^{k-1}) & (3^{k-1}+..) & (3^{k-1}+.. \end{bmatrix}$ But $3^{k-1} + 3^{k-1} + 3^{k-1} = 3 \times 3^{k-1} = 3^k$ $\mathbf{M}^{k+1} = \begin{bmatrix} 3^k & 3^k & 3^k \\ 3^k & 3^k & 3^k \\ 3^k & 3^k & 3^k \end{bmatrix}$ $\therefore \mathbf{M}^{k+1} = \begin{bmatrix} 3^{(k+1)-1} & 3^{(k+1)-1} & 3^{(k+1)-1} \\ 3^{(k+1)-1} & 3^{(k+1)-1} & 3^{(k+1)-1} \\ 3^{(k+1)-1} & 3^{(k+1)-1} & 3^{(k+1)-1} \end{bmatrix}$ $\therefore$P(k + 1) is true Since P(1) is true and P(k) $\Rightarrow$ P(k + 1) hence, by induction, P(n) is true

			for all $n \in N$ **AG**
			Total 5 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
Uses proof by induction and investigates the expression for n = 0 and n = k (must see evidence of both n = 0 and n = k being considered)	AO3.1a	B1	Let $f(n) = 8^n - 7n + 6$ $f(0) = 1 + 6 = 7$ $\Rightarrow f(n)$ is divisible by 7 when $n = 0$ Consider $n = k$
Shows that statement is true for n = 0	AO1.1b	B1	Assume that $f(k)$ is divisible by 7
Commences argument by considering $f(k+1)$ in terms of $f(k)$	AO2.1	R1	$f(k + 1) = 8^{k+1} - 7(k + 1) + 6$
Makes correct deduction that if $f(n)$ is divisible by 7 then $f(n+1)$ is also divisible by 7	AO2.2a	R1	$f(k + 1) - 8f(k) = 56k - 7(k + 1) + 6 - 48$ $f(k + 1) - 8f(k) = 49k - 49$ $f(k + 1) = 8f(k) + 49(k - 1)$ $= 8f(k) + 7(7k - 7)$ $\therefore f(k + 1)$ is divisible by 7 since $f(k)$ is divisible by 7 Therefore $f(k)$ is divisible by 7 $\Rightarrow f(k + 1)$ is divisible by 7
Completes a rigorous argument and explains how their argument proves the required result. **AG**	AO2.4	R1	Since $f(0)$ is divisible by 7 *and* $f(k)$ is divisible by 7 $\Rightarrow f(k + 1)$ is divisible by 7 then, by induction, $f(n) = 8^n - 7n + 6$ is divisible by 7 for all integers $n \geq 0$

ALT				Let $f(n) = 8^n - 7n + 6$ $f(0) = 1 + 6 = 7$ $\Rightarrow f(n)$ is divisible by 7 when n=0

				Consider $n = k$ Assume that $\mathrm{f}(k)$ is divisible by 7 $\mathrm{f}(k + 1)$ $= 8^{k+1} - 7(k + 1) + 6$ $\boldsymbol{8^k - 7k + 6) + 8 \times 7k - 7k - 1 - 48}$ $8\mathrm{f}(k) + 49k - 49$ $8\mathrm{f}(k) + 7(7k - 7)$ $\therefore$ $\mathrm{f}(k+1)$ is divisible by 7 since $\mathrm{f}(k)$ is divisible by 7 Therefore $\mathrm{f}(k)$ is divisible by 7 $\Rightarrow$ $\mathrm{f}(k+1)$ is divisible by 7 Since $\mathrm{f}(0)$ is divisible by 7 *and* $\mathrm{f}(k)$ is divisible by 7 $\Rightarrow$ $\mathrm{f}(k+1)$ is divisible by 7 then, by induction, $\mathrm{f}(n) = 8^n - 7n + 6$ is divisible by 7 for all integers $n \geq 0$
	Total 5 marks			

Q4.

(a) (i) $\mathrm{p}(k + 1) - \mathrm{p}(k) = k^3 + (k + 1)^3 + (k + 2)^3 - (k - 1)^3 - k^3 - (k + 1)^3$

M1

$= (k + 2)^3 - (k - 1)^3$

$= k^3 + 6k^2 + 12k + 8 - (k^2 - 3k^2 + 3k - 1)$

multiplied out & correct unsimplified

A1

$= 9k^2 + 9k + 9 \quad = 9(k^2 + k + 1)$

which is a multiple of 9 (since $k^2 + k + 1$ is an integer)

correct algebra plus statement

A1cso

3

(ii) $\mathrm{p}(1) = 1 + 8 = 9$

$\Rightarrow \mathrm{p}(1)$ is a multiple of 9

result true for n = 1

B1

$p(k + 1) = p(k) + 9(k^2 + k + 1)$
or $p(k+ 1) = p(k) + 9N$

*$p(k + 1)$ = … **and** result from part (i) considered **and** mention of divisible by 9*

M1

Assume $p(k)$ is a multiple of 9

must have word such as "assume" for A1

so $p(k) = 9M$, where M is integer
$\Rightarrow p(k + 1) = 9M + 9N = 9(M + N)$
$\Rightarrow p(k + 1)$ is a multiple of 9

convincingly shown

A1

Result true for $n = 1$ therefore true for $n = 2$, $n = 3$ etc by induction.
(or $p(n)$ is a multiple of 9 for all integers $n \geq 1$)

E1

4

(b) $p(n) = (n - 1)^3 + n^3 + (n + 1)^3$

need to see this OE as evidence

$= 3n^3 + 6n$

or $3n(n^2 + 2)$

B1

$p(n) = 3n(n^2 + 2)$ & $p(n)$ is a multiple of 9.

both of these required

Therefore $n(n^2 + 2)$ is a multiple of 3 (for any positive integer n.)

plus concluding statement

E1

2

[9]

Q5.

$n = 1$, $\frac{3+1}{3-1} = \frac{4}{2} = 2$

be convinced they have used $u_n = \frac{3n+1}{3n-1}$

($u_1 = 2$ so formula is) true when $n = 1$

B1

Assume formula is true for $n = k$ (*)

$(u_{k+1} =)$ $\dfrac{5\dfrac{3k+1}{3k-1}-3}{3\dfrac{3k+1}{3k-1}-1}$

clear attempt at RHS of this formula

M1

clear attempt to remove "double fraction"

m1

$(u_{k+1} =)$ $\dfrac{5(3k+1)-3(3k-1)}{3(3k+1)-(3k-1)}$

$\dfrac{6k+8}{6k+4}$

A1

$(u_{k+1} =)$ $\dfrac{3k+4}{3k+2}$ *or* $u_{k+1} = \dfrac{3(k+1)+1}{3(k+1)-1}$

must have "u_{k+1}=" on at least this line

A1cso

Hence formula is true for $n = k + 1$ (**)

must have lines (*) & (**) **and**
"Result true for n = 1 therefore true for
n = 2, n = 3 etc by induction."

must also have earned previous 5 marks before E1 is scored

E1

[6]

Q6.

Assume result true for $n = k$

Then $u_{k+1} = \dfrac{3}{4-\left(\dfrac{3^{k+1}-3}{3^{k+1}-1}\right)}$

M1

$= \dfrac{3(3^{k+1}-1)}{4(3^{k+1}-1)-(3^{k+1}-3)}$

A1

$4 \times 3^{k+1} - 3^{k+1} = 3^{k+2}$

clearly shown

A1

$u_{k+1} = \dfrac{3^{k+2} - 3}{3^{k+2} - 1}$

A1

$n = 1 \quad \dfrac{3^2 - 3}{3^2 - 1} = \dfrac{3}{4} = u_1$

B1

Induction proof set out properly

must have earned previous 5 marks

E1

[6]

Q7.

(a) Assume true for $n = k$

Then $\displaystyle\sum_{r=1}^{k+1} \frac{2r+1}{r^2(r+1)^2} = 1 - \frac{1}{(k+1)^2} + \frac{2k+3}{(k+1)^2(k+2)^2}$

M1A0 if no LHS

M1A1

$= 1 - \dfrac{1}{(k+1)^2}\left(1 - \dfrac{2k+3}{(k+2)^2}\right)$

attempt to factorise or put over a common denominator

m1

$= 1 - \dfrac{1}{(k+1)^2}\left(\dfrac{k^2+2k+1}{(k+2)^2}\right)$

any correct combination starting 1−

A1

$= 1 - \dfrac{1}{(k+2)^2}$

A1

True for $n = 1$ LHS = RHS = $\dfrac{3}{4}$

B1

Method of induction set out properly

must score all 6 previous marks for this mark

E1

7

(b) $(n + 1)^2 > 10^2$ or $\frac{1}{(n+1)^2} > 10^{-5}$

Condone equals

M1

$n + 1 > 316.2$

$n > 315.2$

$n = 316$

A1

2

[9]

Q8.

(a) $f(k + 1) - 5f(k)$

$= 12^{k+1} + 2 \times 5^k - 5(12^k + 2 \times 5^k - 1)$

M1

$= 12^{k+1} + 2 \times 5^k - 5 \times 12^k - 2 \times 5^k$

for expansion of bracket $2 \times 5^{k-1} = 5^k$ clearly shown

A1

$= 12 \times 12^k - 5 \times 12^k = 7 \times 12^k$

clearly shown

A1

3

(b) Assume $f(k) = M(7)$

Then $f(k + 1) = 5f(k) + M(7)$

Not merely a repetition of part (a)

M1

$= M(7)$

clearly shown

A1

$f(1) = 12 + 2 = 14 = M(7)$

B1

Correct inductive process

(award only if all 3 previous marks earned)

E1 4

[7]

Q9.

(a) Expansion of $(k + 1)(4(k + 1)^2 - 1)$

Any valid method – first step correct

M1

$= 4k^3 + 12k^2 + 11k + 3$

AG

A1 2

(b) Assume true for $n = k$

For $n = k + 1$:

$$\sum_{r=1}^{k+1}(2r-1)^2 = \frac{1}{3}k(4k^2-1) + (2k+1)^2$$

No LHS M1A0

M1A1

$$= \frac{1}{3}(4k^3 + 12k^2 + 11k + 3)$$

ft error in (2k + 1)

A1F

$$= \frac{1}{3}(k+1)(4(k+1)^2 - 1)$$

Using part (a)

A1

True for $n = 1$ shown

B1

Proof by induction set out properly

Dependent on all marks correct

E1 6

(if factorised by 3 linear factors, allow A1 at this particular point)

[8]

Q10.

(a) Assume true for $n = k$

$u_{k+1} = 2(3 \times 2^{k-1} - 1) + 1$

M1A1

$= 3 \times 2^k - 1$

$2^{(k-1)+1}$ not necessarily seen

A1

True for $n = 1$ shown

B1

Method of induction clearly expressed

Provided all 4 previous marks earned

E1

5

(b) $$\sum_{r=1}^{n} u_r = \sum_{r=1}^{n} 3 \times 2^{r-1} - n$$

$= 3(2^n - 1) - n$

M1 for summation, ie recognition of a GP

M1A1

$= u_{n+1} - (n + 2)$

AG

A1

3

[8]

Q11.

(a) $$\frac{1}{(k+2)!} = \frac{k+3}{(k+3)!}$$

M1

Result

A1

2

(b) Assume true for $n = k$

For $n = k + 1$

$$\sum_{r=1}^{k+1} \frac{r \times 2^r}{(r+2)!} = 1 - \frac{2^{k+1}}{(k+2)!} + \frac{(k+1)2^{k+1}}{(k+3)!}$$

If no LHS of equation, M1A0

M1 A1

$$= 1 - 2^{k+1}\left(\frac{1}{(k+2)!} - \frac{k+1}{(k+3)!}\right)$$

m1 for a suitable combination clearly shown

m1

$$= 1 - \frac{2^{k+2}}{(k+3)!}$$

clearly shown or stated true for n = k + 1

A1

True for n = 1

Shown

B1

Method of induction set out properly

Provided previous 5 marks all earned

E1

6

[8]

Q12.

Assume result true for $n = k$

Then $\sum_{r=1}^{k+1} \frac{2^r \times r}{(r+1)(r+2)}$

SC *If no series at all indicated on LHS, deduct 1 and give E0 at end*

$$= \frac{2^{k+1}}{k+2} + \frac{2^{k+1}(k+1)}{(k+2)(k+3)} - 1$$

M1A1

$$= \frac{2^{k+1}(k+3+k+1)}{(k+2)(k+3)} - 1$$

A1

$$= \frac{2^{k+1}2(k+2)}{(k+2)(k+3)} - 1$$

Putting over common denominator (not including the –1, unless separated later)

M1

$= \frac{2^{k+2}}{k+3} - 1$

A1

$k = 1$: LHS $= \frac{1}{3}$, RHS $= \frac{2^2}{3} - 1$

B1

$P_k \Rightarrow P_{k+1}$ and P_1 true

Must be completely correct

E1

[7]

Q13.

(a) $(\cos\theta + i\sin\theta)^{k+1} =$
$(\cos k\theta + i\sin k\theta)(\cos\theta + i\sin\theta)$

M1

Multiply out

Any form

A1

$= \cos(k+1)\theta + i\sin(k+1)\theta$

Clearly shown

A1

True for $n = 1$ shown

B1

$P(k) \Rightarrow P(k+1)$ and $P(1)$ true

provided previous 4 marks earned

E1 5

(b) $\frac{1}{z^n} = \frac{1}{\cos n\theta + i\sin n\theta} = \cos n\theta - i\sin n\theta$

or $z^{-n} = \cos(-n\theta) + i\sin(-n\theta)$

$(\cos\theta + i\sin\theta)^{-n}$

SC quoted as $\cos n\theta - i\sin n\theta$

earns M1A1 only

M1A1

$$z^n + \frac{1}{z^n} = 2\cos n\theta$$

AG

A1

3

(c) $z + \frac{1}{z} = \sqrt{2}$

$2\cos\theta = \sqrt{2}$

M1

$\theta = \frac{\pi}{4}$

A1

$z^{10} + \frac{1}{z^{10}} = 2\cos\left(\frac{10\pi}{4}\right)$

M0 for merely writing

$z^{10} + \frac{1}{z^{10}} = 2\cos 10\theta$

M1

= 0

A1F

4

[12]

Q14.

Assume result true for $n = k$

Then $\sum_{r=1}^{k+1} (r^2 + 1)r!$

$= ((k + 1)^2 + 1)(k + 1)! + k(k + 1)!$

M1A1

Taking out $(k + 1)!$ as factor

m1

$= (k + 1)!(k^2 + 2k + 1 + 1 + k)$

A1

$= (k + 1)(k + 2)!$

A1

$k = 1$ shown $(1^2 + 1)1! = 2$

$1 \times 2! = 2$

B1

$P_k \Rightarrow P_{k+1}$ and P_1 true

If all 6 marks earned

E1

[7]

Q15.

(a) Clear reason given

Minimum O × E = E

E1

1

(b) (i) $(k + 1)((k + 1)^2 + 5) – k(k^2 + 5)$

M1

$= 3k^2 + 3k + 6$

A1

$k^2 + k = k(k + 1) = M(2)$

***Must** be shown*

E1

$f(k + 1) – f(k) = M(6)$

E1

4

(ii) Assume true for $n = k$

$f(k + 1) – f(k) = M(6)$

Clear method

M1

$\therefore\ f(k + 1) = M(6) + f(k)$

$= M(6) + M(6)$

A1

$= M(6)$

True for $n = 1$

B1

$P(n) \rightarrow P(n + 1)$ and $P(1)$ true

Provided all other marks earned in (b)(ii)

E1

4

[9]

Q16.

(a) Assume true for $n = k$

$(\cos\theta + i\sin\theta)^{k+1}$

$= (\cos k\theta + i\sin k\theta)(\cos\theta + i\sin\theta)$

M1

Multiply out

Any form

A1

$= \cos(k+1)\theta + i\sin(k+1)\theta$

A1

True for $n = 1$ shown

B1

$P(k) \Rightarrow P(k + 1)$ and $P(1)$ true

Allow E1 only if previous 4 marks earned

E1

5

(b) $\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)^6 = \cos\frac{6\pi}{6} + i\sin\frac{6\pi}{6}$

M1

$= -1$

A1

2

(c) $(\cos\theta + i\sin\theta)(1 + \cos\theta - i\sin\theta)$

M1

$= \cos\theta + \cos^2\theta - i\sin\theta\cos\theta + i\sin\theta + i\sin\theta\cos\theta + \sin^2\theta$

(Accept $-i^2 \sin^2\theta$)

Or $e^{i\theta}(1 + e^{-i\theta})$

A1

$= 1 + \cos\theta + i\sin\theta$

AG

A1 3

(d) $\theta = \frac{\pi}{6}$ used

In the context of part (c)

M1

Part (c) raised to power 6

M1

Use of result in part (b)

A1

$\left(1+\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)^6 + \left(1+\cos\frac{\pi}{6}-i\sin\frac{\pi}{6}\right)^6 = 0$

AG

A1 4

[14]

Q17.

(a) $\left(1-\frac{1}{(k+1)^2}\right)\times\frac{k+1}{2k}=\frac{(k+1)^2-1}{(k+1)^2}\times\frac{k+1}{2k}$

M1

$=\frac{k^2+2k}{(k+1)^2}\times\frac{k+1}{2k}$

A1

$=\frac{k+2}{2(k+1)}$

AG

A1 3

(b) Assume true for $n = k$, then

$\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\ldots\left(1-\frac{1}{(k+1)^2}\right)$

M1

$= \frac{k+2}{2(k+1)}$

A1

True for n = 2 shown $1 - \frac{1}{2^2} = \frac{3}{4}$

B1

$P_n \Rightarrow P_{n+1}$ and P_2 true

only if the other 3 marks earned

E1

4

[7]

Q18.

(a) Assume result true for $n = k$

$\sum_{r=1}^{k}(r+1)2^{r-1} = k2^k$

$\sum_{r=1}^{k+1}(r+1)2^{r-1} = k2^k + (k+2)2^k$

M1A1

$= 2^k(k + k + 2)$

m1

$= 2^k(2k + 2)$

$= 2^{k+1}(k + 1)$

A1

$n = 1$ $\quad 2 \times 2^0 = 2 = 1 \times 2^1$

B1

$P^k \Rightarrow P^k + 1$ and P_1 is true

Provided previous 5 marks earned

E1

6

(b) $\sum_{r=1}^{2n}(r+1)2^{r-1} - \sum_{r=1}^{n}(r+1)2^{r-1}$

Sensible attempt at the difference between 2 series

M1

$= 2n\, 2^{2n} - n2^n$

A1

$= n(2^{n+1} - 1)2^n$

AG

A1

3

[9]

Q19.

(a) $f(n + 1) - 8f(n) = 15^{n+1} - 8^{n-1}$

$- 8(15^n - 8^{n-2})$

M1A1

$= 15^{n+1} - 8.15^n$

$= 15^n (15 - 8)$

For multiples of powers of 15 only

M1

$= 7.15^n$

For valid method ie not using 120^n etc

A1

4

(b) *Assume f(n) is M(7)*

Then $f(n + 1) - 8f(n) = 7 \times 15^n$

Or considering $f(n + 1) - f(n)$

M1

$f(n + 1) = M(7) + M(7)$
$= M(7)$

A1

$n = 2$: $f(n) = 15^2 - 8^0 = 224$
$= 7 \times 32$

B1

$n = 1$ B0

$P(n) \Rightarrow P(n + 1)$ and $P(2)$ true

Must score previous 3 marks to be awarded

E1

4

[8]

EXAM PAPERS PRACTICE

Examiner reports

Q1.

The vast majority of students clearly knew how to structure a proof by induction, but only a third of the students could produce a fully correct proof. Almost all attempts included a confirmation of the initial case of $n = 1$, and many of these then assumed the identity was true for $n = k$. Good students were able to successfully manipulate this equation into a form which satisfied the $n = k + 1$ case. Some of these, however, resorted to expanding their quartic expression and then demonstrating that it was equivalent to the required quartic. Although an acceptable method, this was a time consuming approach, and prone to error. However, those who succeeded in this usually went on to produce a satisfactory proof by induction, although a minority demonstrated a lack of understanding of the requirements of a proof by induction.

The majority of students were able to make some progress in part (b), correctly substituting $2n$ into the appropriate formulae, but some of these then struggled with the algebra. Again, a minority of students opted to expand their quartic, instead of factorising, but then errors often crept in.

Q4.

(a) (i) *The algebra was handled quite well here, but having obtained a simplified expression such as $9(k^2 + k + 1)$, it was necessary to make a statement that the expression was divisible by 9.*

(ii) *Very few students scored full marks for the proof by induction. Most students were confused as to what they were trying to prove. A large number of students' attempts contained expressions such as "$M9$" without defining their notation. Some even wrote things like "$M9 + M9 = M18$". Examiners were often not convinced that students were actually trying to show that $p(k + 1)$ was divisible by 9, particularly when they omitted key phrases such as "assume that $p(k)$ is a multiple of 9". Even the mark for the case when $n = 1$ was rarely earned since students omitted to write a conclusion such as "therefore $p(1)$ is a multiple of 9" after showing that $p(1) = 9$.*

Many students seemed only familiar with proof by induction involving formulae for sums of series as they wrote things such as "adding the $(k + 1)$th term" or "LHS= …" and "RHS= …" or "formula is true" with no reference to divisibility at all.

(b) *It was necessary for students to show that $p(n) = (n - 1)^3 + n^3 + (n + 1)^3$ could be simplified to $3n^3 + 6n$ in order to* *convince examiners that they were not working back from the printed answer. Clearly the simplification of p(n) may have appeared in an earlier part of the question, in which case credit was given for simply writing p(n) $3n^3 + 6n$ here. Most students who found this expression for p(n) legitimately were able to deduce that since p(n) was a multiple of 9 then the printed expression was a multiple of 3.*

Q5.

Some students had been well taught in the setting out of proofs by induction; others sadly floundered as if they were trying to remember the words of a song but could only hum the tune. Several students tried **adding** the next term instead of realizing that this proof involved the nth term of a sequence. Quite a few weaker students, who realised what they wanted to achieve, were defeated by the double quotient and could not cope with the algebra. Many who wrote $\frac{3+1}{3-1} = \frac{4}{2}$ = 2 failed to make a further statement such as "the formula is true when $n = 1$" . Those students who use expressions such as "$P(k)$ is true" are well advised to define what they mean by the proposition P(k). Several students who set out their proof quite well spoiled their solution by writing "therefore true for all $n \geq 1$ ".

Q6.

Responses to this question were mixed. There were some excellent answers with candidates displaying clear knowledge and presentation of the method of induction, but there were also some very muddled answers in which sequences were mixed with series, and these solutions showed candidates trying to add u_k to u_{k+1}.
There was also some poor algebra in the substitution for u_k in the formula for u^{k+1} with expressions such as $4 \times 3^{k+1}$ being written as 12^{k+1}.

Q7.

Whilst most students made an attempt at this question, not many provided a convincing solution. The mechanics of the method of induction were usually present, but a real understanding of the theory and argument behind it was clearly lacking on many scripts. In showing that the sum to $k + 1$ terms of the given series was equal to the sum to k terms plus the $(k + 1)^{th}$ term students made clear sign errors with consequent attempts to make the working out fit the solution. Part (b) was less successful, especially when a student used inequalities. Whilst the use of equalities was condoned, some students having arrived at 315.2, still gave a final incorrect answer.

Q8.

Responses to this question were very mixed. Whilst in part (a) candidates knew what to do, after writing down a correct expression for f(k + 1) – f(k) terms such as 60^{k-1} appeared quite frequently. In part (b), there was a lot of muddled logic, with many candidates thinking that the reiteration of part (a) was a sufficient argument to prove part (b). There was also clear evidence of a lack of knowledge regarding how proofs by induction should be set out.

Q9.

Virtually all candidates completed part (a) successfully. However, in part (b), candidates generally were divided into three classes. There were those who produced a first class proof by induction. Then there were those who would have produced a good solution had their algebra not gone askew in the algebraic part of their proof. This was often due to not recognising the hint given in part (a). Finally, there were those who knew the mechanics of a proof by induction, but without a true understanding of the principles involved. To give an example of this, it was quite common to see the sum to k terms of the series added to the $(k + 1)^{th}$ term without any reference to what the sum of those two terms represented. Explanations generally were poor in this category.

Q10.

Candidates generally had difficulty in using the recurrence relationship in their proof by induction, so that responses to this question were rather poor. Proper detail is essential in the proof by induction using a sequence and a sequence relationship so that relatively few candidates scored full marks for part (a).

Very few candidates indeed were successful in part (b). Only a handful of candidates recognised that a Geometric Progression was involved. If they did, they usually went on to obtain a correct solution. It should perhaps be added that one method of providing an excellent solution was to rewrite $u_{k+1} = 2u_k + 1$ as $u_{k+1} - u_k = u_k + 1$ and then to use the method of differences to sum the series; but this method of solution was extremely rare.

Q11.

Again responses to this question were mixed. It was evident that some candidates thought that $(k + 2)!$ started at k, and wrote it as $k(k + 1)(k + 2)$. Others wrote down the result after some rather dubious algebra. Although there has been considerable improvement in the way that solutions by induction have been expressed, in this case what would otherwise have been acceptable solutions were spoilt by errors of sign. The same error occurred frequently. It occurred when a candidate tried to combine, in one bracket, a negative expression followed by a positive expression by placing a negative sign outside the combining bracket and then by forgetting to alter the sign before the positive term to compensate.

Q12.

One thing which became evident in the marking of this question was that although candidates were able to perform the mechanics of proof by induction they did not really understand the theory behind it. In a significant number of solutions not one reference to a series or the use of the Σ symbol occurred. Solutions started 'assume result true for $n = k$' followed by $\frac{2^{k+1}}{n+2} - 1 + \frac{2^{k+1} \times (k+1)}{(k+2)(k+3)}$ which was duly shown to be $\frac{2^{k+2}}{k+3} - 1$.

Q13.

The general rules applicable to proof by induction in part (a) were usually understood, but because candidates realised that the product of $\cos\theta + i\sin\theta$ with $\cos k\theta + i\sin k\theta$ had to result in $\cos(k + 1)\theta + i\sin(k + 1)\theta$, many lost marks through omitting some of the intermediate steps.

In part (b), many candidates lost a mark by assuming that $(\cos\theta + i\sin\theta)^{-n}$ was equal to $\cos n\theta - i\sin n\theta$ without any justification.

Part (c) was almost invariably correctly done.

Q14.

Responses to this question varied considerably. It was not, in general, that candidates did not understand the method of induction but rather that the algebraic manipulation especially in the handling of factorials proved to be a stumbling block. For instance $k(k+1)!$ would be written as $(k^2+k)!$ and in a significant number of solutions candidates, having managed to reach $(k+1)(k+2)(k+1)!$, abandoned their solutions not realising that the result was, in fact, $(k+1)(k+2)!$.

Q15.

There were many good responses to part (a) and to part (b)(i) of this question, with the majority of candidates arriving at $3k^2 + 3k + 6$ and many going on correctly to show that this expression was a multiple of 6. However, some candidates wrote the expression as

$\frac{1}{2}.6(k^2+k)+6$ and claimed from this rewritten form that the expression was a multiple of 6. Solutions to part (b)(ii) were somewhat disappointing with many candidates failing to convince that $f(k+1)$ was a multiple of 6 if $f(k)$ was, and failing to realise that the formal method of induction requires crystal clarity.

Q16.

It was clear that a good number of candidates had not met the proof of de Moivre's Theorem by induction before and there were not many solutions gaining full marks. It was also not uncommon to see expressions such as

$\cos k\theta + i\sin k\theta + \cos\theta + i\sin\theta = \cos(k+1)\theta + i\sin(k+1)\theta$.

Parts (b) and (c) were generally well done, although in part (c) a number of candidates, when multiplying $i\sin\theta$ by $-i\sin\theta$, wrote $-i\sin^2\theta$ and thus were unable to complete this part satisfactorily. Those candidates who spotted the connection between parts (c) and (d) usually went on to write out a correct solution to part (d), but it was disappointing to see

$\left(1+\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)^6$ written as $i^6+\cos\frac{6\pi}{6}+i\sin\frac{6\pi}{6}$ with alarming regularity.

Q17.

Part (a) was usually answered correctly although there were many very long-winded algebraic methods employed including the multiplication out of just about every bracket followed immediately by their re-factorisation. There was however much muddled thinking in part (b). Whilst most candidates had some outline of the method of induction many candidates attempted this part with no reference whatever to the series product in question, whilst others tried to **add** the $(k+1)^{\text{th}}$ term to the sum of k products. Candidates who did consider the series usually used $\sum$ rather than Π but this was not penalised.

Q18.

Responses to this question were only fair. Although candidates had some idea of what was required for the inductive process, in part (a) they appeared to be easily confused. Common statements were for instance $(k+2)\,2^k = k2^k$ or $(k+1)^{2k-1} + (k+2)\,2^k$ or to even write down correctly $k2^k + (k+2)2^k$ but without any reference whatsoever as to what the expression represented. In part (b), unless candidates realised that the given series was the difference of two other series no progress was made and only a few realised the connection with part (a). A common approach was to try and prove this result by induction also.

Q19.

Although there were some good solutions to part(a) of this question it did show in many cases a lack of understanding of the theory of indices. It was quite common to see 8×15^n written as a 120" and $8 x 8^{n-2}$as 64^{n-2}. There was also a lack of clarity in part (b). It was not unusual to see the first line of the inductive proof to state "Assume result true for $n = k$ i.e. that f (k) = $15^k - 8^{fc-2}$" to be followed by "f (k + 1) 8f (k)is a multiple of 7", showing a lack of understanding of the proof by induction in the case of multiples of integers. Some candidates tried to establish the result for n = 1 in spite of being told that n was greater than or equal to 2. A substantial minority of candidates ignored the hint in part (a) and in part (b) considered f (k + 1) – f (k) with a measure of success.